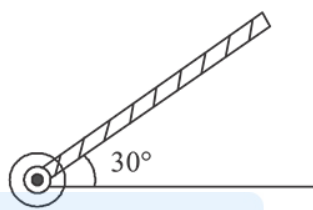


JEE Mains 2019 Chapter wise Question Bank

Rotational Motion - Questions

Q1

A rod of length 50 cm is pivoted at one end. It is raised such that it makes an angle of 30° from the horizontal as shown and released from rest. Its angular speed when it passes through the horizontal (in rads^{-1}) will be ($g = 10 \text{ ms}^{-2}$)



- (1) $\sqrt{\frac{30}{7}}$ (2) $\sqrt{30}$
 (3) $\sqrt{\frac{20}{3}}$ (4) $\sqrt{\frac{30}{2}}$

9 Jan Evening

Q2

A mass of 10 kg is suspended vertically by a rope from the roof. When a horizontal force is applied on the rope at some point, the rope deviated at an angle of 45° at the roof point. If the suspended mass is at equilibrium, the magnitude of the force applied is ($g = 10 \text{ ms}^{-2}$)

- (1) 200 N (2) 140 N
 (3) 70 N (4) 100 N

9 Jan Evening

Q3

To mop-clean a floor, a cleaning machine presses a circular mop of radius R vertically down with a total force F and rotates it with a constant angular speed about its axis. If the force F is distributed uniformly over the mop and if coefficient of friction between the mop and the floor is μ , the torque, applied by the machine on the mop is:

- (1) $\mu FR/3$ (2) $\mu FR/6$
 (3) $\mu FR/2$ (4) $\frac{2}{3} \mu FR$

10 Jan Morning

Q4

A homogeneous solid cylindrical roller of radius R and mass M is pulled on a cricket pitch by a horizontal force. Assuming rolling without slipping, angular acceleration of the cylinder is:

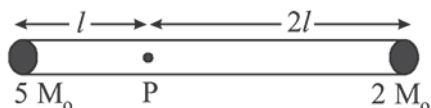
- (1) $\frac{3F}{2mR}$ (2) $\frac{F}{3mR}$
 (3) $\frac{F}{2mR}$ (4) $\frac{2F}{3mR}$

10 Jan Morning

Q5

Rotational Motion

A rigid massless rod of length $3l$ has two masses attached at each end as shown in the figure. The rod is pivoted at point P on the horizontal axis (see figure). When released from initial horizontal position, its instantaneous angular acceleration will be:

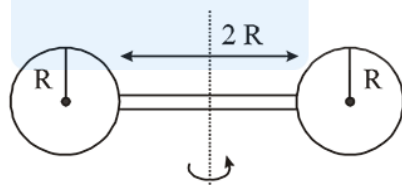


- (1) $\frac{g}{13l}$ (2) $\frac{g}{3l}$
 (3) $\frac{g}{2l}$ (4) $\frac{7g}{3l}$

10 Jan Evening

Q6

Two identical spherical balls of mass M and radius R each are stuck on two ends of a rod of length $2R$ and mass M (see figure). The moment of inertia of the system about the axis passing perpendicularly through the centre of the rod is:



- (1) $\frac{137}{15}MR^2$ (2) $\frac{17}{15}MR^2$
 (3) $\frac{209}{15}MR^2$ (4) $\frac{152}{15}MR^2$

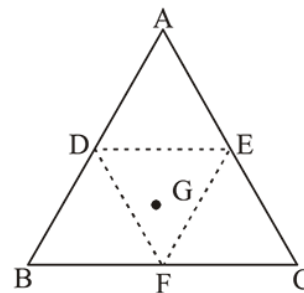
10 Jan Evening

Q7

JEE Mains 2019 Chapter wise Question Bank

An equilateral triangle ABC is cut from a thin solid sheet of wood. (See figure) D, E and F are the mid-points of its sides as shown and G is the centre of the triangle. The moment of inertia of the triangle about an axis passing through G and perpendicular to the plane of the triangle is I_0 . If the smaller triangle DEF is removed from ABC, the moment of inertia of the remaining figure about the same axis is I.

Then :

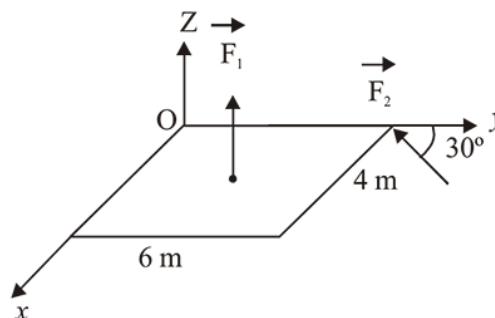


- (1) $I = \frac{15}{16}I_0$ (2) $I = \frac{3}{4}I_0$
 (3) $I = \frac{9}{16}I_0$ (4) $I = \frac{I_0}{4}$

11 Jan Morning

Q8

A slab is subjected to two forces $\vec{F}_1$ and $\vec{F}_2$ of same magnitude F as shown in the figure. Force $\vec{F}_2$ is in XY-plane while force $\vec{F}_1$ acts along z-axis at the point $(2\vec{i} + 3\vec{j})$. The moment of these forces about point O will be :



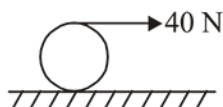
- (1) $(3\hat{i} - 2\hat{j} + 3\hat{k})F$ (2) $(3\hat{i} - 2\hat{j} - 3\hat{k})F$
 (3) $(3\hat{i} + 2\hat{j} - 3\hat{k})F$ (4) $(3\hat{i} + 2\hat{j} + 3\hat{k})F$

11 Jan Morning

Q9

Rotational Motion

a string is wound around a hollow cylinder of mass 5 kg and radius 0.5 m. If the string is now pulled with a horizontal force of 40 N, and the cylinder is rolling without slipping on a horizontal surface (see figure), then the angular acceleration of the cylinder will be (Neglect the mass and thickness of the string) :



- (1) 20 rad/s^2 (2) 16 rad/s^2
(3) 12 rad/s^2 (4) 10 rad/s^2

11 Jan Evening

Q10

The magnitude of torque on a particle of mass 1 kg is 2.5 Nm about the origin. If the force acting on it is 1 N, and the distance of the particle from the origin is 5m, the angle between the force and the position vector is (in radians):

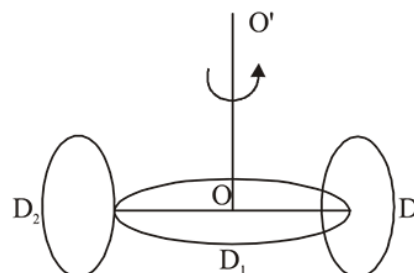
- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3}$
(3) $\frac{\pi}{8}$ (4) $\frac{\pi}{4}$

11 Jan Evening

Q11

JEE Mains 2019 Chapter wise Question Bank

A circular disc D_1 of mass M and radius R has two identical discs D_2 and D_3 of the same mass M and radius R attached rigidly at its opposite ends (see figure). The moment of inertia of the system about the axis OO' , passing through the centre of D_1 , as shown in the figure, will be :



- (1) MR^2 (2) $3MR^2$
(3) $\frac{4}{5}MR^2$ (4) $\frac{2}{3}MR^2$

11 Jan Evening

Q12

Let the moment of inertia of a hollow cylinder of length 30 cm (inner radius 10 cm and outer radius 20 cm), about its axis be I . The radius of a thin cylinder of the same mass such that its moment of inertia about its axis is also I , is:

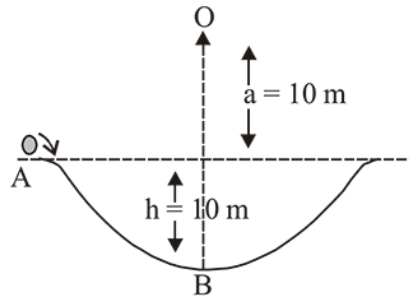
- (1) 12 cm (2) 16 cm
(3) 14 cm (4) 18 cm

12 Jan Morning

Q13

Rotational Motion

A particle of mass 20 g is released with an initial velocity 5 m/s along the curve from the point A, as shown in the figure. The point A is at height h from point B. The particle slides along the frictionless surface. When the particle reaches point B, its angular momentum about O will be : (Take $g = 10 \text{ m/s}^2$)

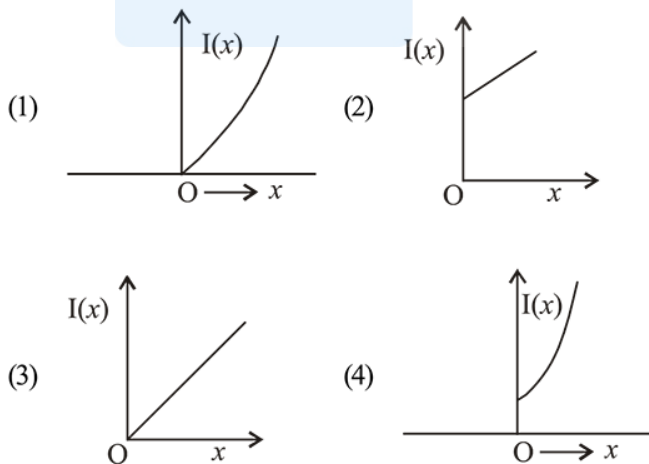


- (1) $2 \text{ kg-m}^2/\text{s}$ (2) $8 \text{ kg-m}^2/\text{s}$
(3) $6 \text{ kg-m}^2/\text{s}$ (4) $3 \text{ kg-m}^2/\text{s}$

12 Jan Evening

Q14

The moment of inertia of a solid sphere, about an axis parallel to its diameter and at a distance of x from it, is ' $I(x)$ '. Which one of the graphs represents the variation of $I(x)$ with x correctly?

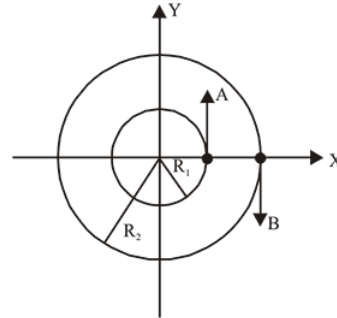


12 Jan Evening

Q15

JEE Mains 2019 Chapter wise Question Bank

Two particles A, B are moving on two concentric circles of radii R_1 and R_2 with equal angular speed ω . At $t = 0$, their positions and direction of motion are shown in the figure :



The relative velocity $\vec{v}_A - \vec{v}_B$ and $t = \frac{\pi}{2\omega}$ is given by:

- (1) $\omega(R_1 + R_2)\hat{i}$ (2) $-\omega(R_1 + R_2)\hat{i}$
(3) $\omega(R_2 - R_1)\hat{i}$ (4) $\omega(R_1 - R_2)\hat{i}$

12 Jan Evening

Q16

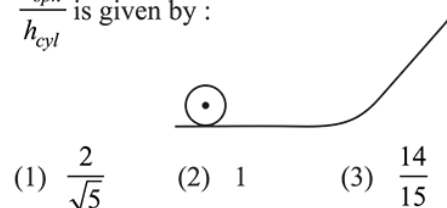
A thin circular plate of mass M and radius R has its density varying as $\rho(r) = \rho_0 r$ with ρ_0 as constant and r is the distance from its center. The moment of Inertia of the circular plate about an axis perpendicular to the plate and passing through its edge is $I = a MR^2$. The value of the coefficient a is:

- (1) $\frac{1}{2}$ (2) $\frac{3}{5}$ (3) $\frac{8}{5}$ (4) $\frac{3}{2}$

8 April Morning

Q17

A solid sphere and solid cylinder of identical radii approach an incline with the same linear velocity (see figure). Both roll without slipping all throughout. The two climb maximum heights h_{sph} and h_{cyl} on the incline. The ratio $\frac{h_{sph}}{h_{cyl}}$ is given by :



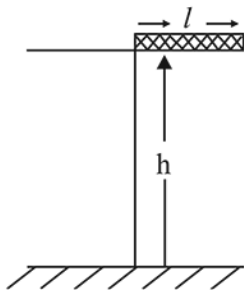
- (1) $\frac{2}{\sqrt{5}}$ (2) 1 (3) $\frac{14}{15}$ (4) $\frac{4}{5}$

8 April Evening

Q18

Rotational Motion

A rectangular solid box of length 0.3 m is held horizontally with one of its sides on the edge of a platform of height 5m. When released, it slips off the table in a very short time $\tau = 0.01$ s, remaining essentially horizontal. The angle by which it would rotate when it hits the ground will be (in radians) close to :



- (1) 0.5 (2) 0.3 (3) 0.02 (4) 0.28

8 April Evening

Q19

A stationary horizontal disc is free to rotate about its axis. When a torque is applied on it, its kinetic energy as a function of θ , where θ is the angle by which it has rotated, is given as $k\theta^2$. If its moment of inertia is I then the angular acceleration of the disc is

- (1) $\frac{k}{4I}\theta$ (2) $\frac{k}{I}\theta$ (3) $\frac{k}{2I}\theta$ (4) $\frac{2k}{I}\theta$

9 April Morning

Q20

The following bodies are made to roll up (without slipping) the same inclined plane from a horizontal plane : (i) a ring of radius R , (ii) a solid cylinder of radius $\frac{R}{2}$ and (iii) a solid sphere of radius $\frac{R}{4}$. If, in each case, the speed of the center of mass at the bottom of the incline is same, the ratio of the maximum heights they climb is :

- (1) 4 : 3 : 2 (2) 10 : 15 : 7
(3) 14 : 15 : 20 (4) 2 : 3 : 4

9 April Morning

Q21

Moment of inertia of a body about a given axis is 1.5 kg m^2 . Initially the body is at rest. In order to produce a rotational kinetic energy of 1200 J, the angular acceleration of 20 rad/s^2 must be applied about the axis for a duration of:

- (1) 2.5s (2) 2s (3) 5s (4) 3s

JEE Mains 2019 Chapter wise Question Bank

9 April Evening

Q22

A thin smooth rod of length L and mass M is rotating freely with angular speed ω_0 about an axis perpendicular to the rod and passing through its center. Two beads of mass m and negligible size are at the center of the rod initially. The beads are free to slide along the rod. The angular speed of the system, when the beads reach the opposite ends of the rod, will be:

- (1) $\frac{M\omega_0}{M+m}$ (2) $\frac{M\omega_0}{M+3m}$
(3) $\frac{M\omega_0}{M+6m}$ (4) $\frac{M\omega_0}{M+2m}$

9 April Evening

Q23

A particle of mass m is moving along a trajectory given by $x = x_0 + a \cos \omega_1 t$

$$y = y_0 + b \cos \omega_2 t$$

The torque, acting on the particle about the origin, at $t = 0$ is :

- (1) $m(-x_0 b + y_0 a) \omega_1^2 \hat{k}$
(2) $+m y_0 a \omega_1^2 \hat{k}$
(3) zero
(4) $-m(x_0 b \omega_2^2 - y_0 a \omega_1^2) \hat{k}$

10 April Morning

Q24

A thin disc of mass M and radius R has mass per unit area $\sigma(r) = kr^2$ where r is the distance from its centre. Its moment of inertia about an axis going through its centre of mass and perpendicular to its plane is :

- (1) $\frac{MR^2}{3}$ (2) $\frac{2MR^2}{3}$ (3) $\frac{MR^2}{6}$ (4) $\frac{MR^2}{2}$

10 April Morning

Q25

Rotational Motion

Two coaxial discs, having moments of inertia I_1 and $\frac{I_1}{2}$, are rotating with respective angular velocities ω_1 and $\frac{\omega_1}{2}$, about their common axis. They are brought in contact with each other and thereafter they rotate with a common angular velocity. If E_f and E_i are the final and initial total energies, then $(E_f - E_i)$ is :

- (1) $-\frac{I_1\omega_1^2}{12}$ (2) $\frac{I_1\omega_1^2}{6}$ (3) $\frac{3}{8}I_1\omega_1^2$ (4) $-\frac{I_1\omega_1^2}{24}$

10 April Morning

Q26

A solid sphere of mass M and radius R is divided into two unequal parts. The first part has a mass of $\frac{7M}{8}$ and is converted into a uniform disc of radius $2R$. The second part is converted into a uniform solid sphere. Let I_1 be the moment of inertia of the new sphere about its axis. The ratio I_1/I_2 is given by :

- (1) 185 (2) 140 (3) 285 (4) 65

10 April Evening

Q27

The time dependence of the position of a particle of mass $m = 2$ is given by $\vec{r}(t) = 2t\hat{i} - 3t^2\hat{j}$. Its angular momentum, with respect to the origin, at time $t = 2$ is :

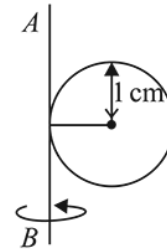
- (1) $48(\hat{i} + \hat{j})$ (2) $36\hat{k}$
(3) $-34(\hat{k} - \hat{i})$ (4) $-48\hat{k}$

10 April Evening

Q28

JEE Mains 2019 Chapter wise Question Bank

A metal coin of mass 5 g and radius 1 cm is fixed to a thin stick AB of negligible mass as shown in the figure. The system is initially at rest. The constant torque, that will make the system rotate about AB at 25 rotations per second in 5s, is close to :



- (1) 4.0×10^{-6} Nm (2) 1.6×10^{-5} Nm
(3) 7.9×10^{-6} Nm (4) 2.0×10^{-5} Nm

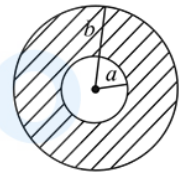
10 April Evening

Q29

A circular disc of radius b has a hole of radius a at its centre (see figure). If the mass per unit area of the disc varies as $\left(\frac{\sigma_0}{r}\right)$, then the radius of gyration of the disc about its axis

passing through the centre is :

- (1) $\sqrt{\frac{a^2 + b^2 + ab}{2}}$ (2) $\frac{a+b}{2}$
(3) $\sqrt{\frac{a^2 + b^2 + ab}{3}}$ (4) $\frac{a+b}{3}$



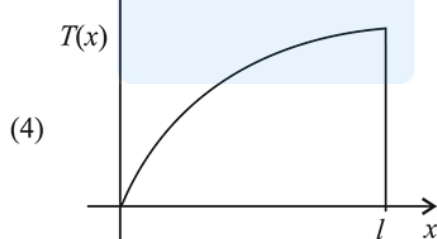
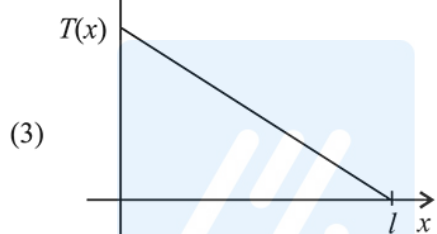
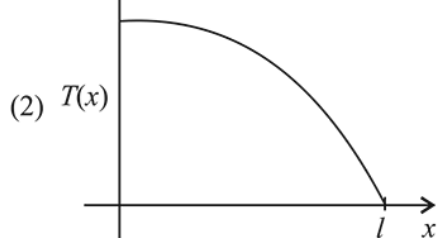
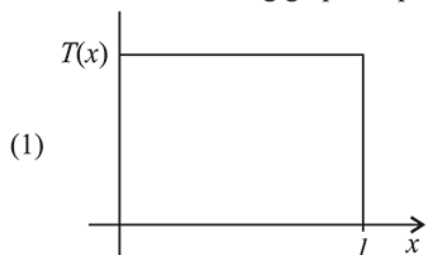
12 April Morning

Q30

Rotational Motion

JEE Mains 2019 Chapter wise Question Bank

A uniform rod of length l is being rotated in a horizontal plane with a constant angular speed about an axis passing through one of its ends. If the tension generated in the rod due to rotation is $T(x)$ at a distance x from the axis, then which of the following graphs depicts it most closely ?



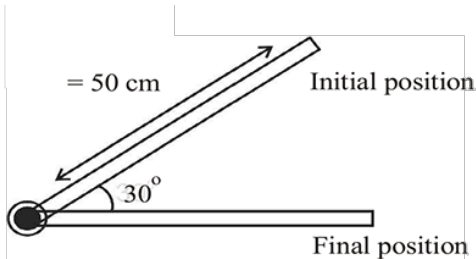
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Rotational Motion - Answers

Q1

(4)



By the law of conservation of energy,
P.E. of rod = Rotational K.E.

$$mg \frac{\ell}{2} \sin \theta = \frac{1}{2} I \omega^2$$

$$\Rightarrow mg \frac{\ell}{2} \sin 30^\circ = \frac{1}{2} \frac{m \ell^2}{3} \omega^2 \Rightarrow mg \frac{\ell}{2} \times \frac{1}{2} = \frac{1}{2} \frac{m \ell^2}{3} \omega^2$$

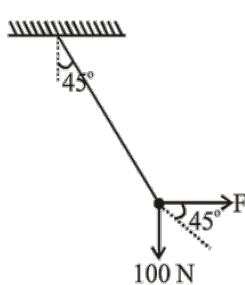
For complete length of rod,

$$\omega = \sqrt{3g/2(2\ell)} = \sqrt{\frac{30}{2}} \text{ rods}^{-1}$$

9 Jan Evening

Q2

(4)



At equilibrium,

$$\tan 45^\circ = \frac{mg}{F} = \frac{100}{F}$$

$$\therefore F = 100 \text{ N}$$

9 Jan Evening

Q3

(4) Consider a strip of radius x and thickness dx ,

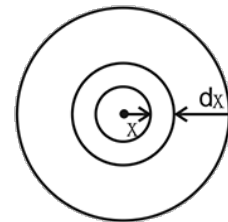
Torque due to friction on this strip

Net torque = $\sum$ Torque on ring

$$\int d\tau = \int_0^R \frac{\mu F \cdot 2\pi x dx}{\pi R^2}$$

$$\Rightarrow \tau = \frac{2\mu F R^3}{R^2 \cdot 3}$$

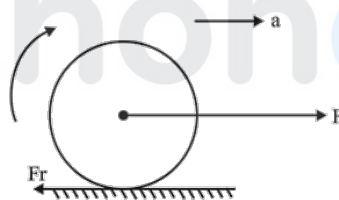
$$\tau = \frac{2\mu FR}{3}$$



10 Jan Morning

Q4

(4)



$$F - f_r = ma \quad \dots(i)$$

$$f_r R = I\alpha = \frac{mR^2}{2} \alpha \quad \dots(ii)$$

for pure rolling

$$a = \alpha R \quad \dots(iii)$$

from (i) (ii) and (iii)

$$F - \frac{mR\alpha}{2} = m\alpha R$$

$$F = \frac{3}{2} mR\alpha$$

$$\alpha = \frac{2F}{3mR}$$

10 Jan Morning

Q5

Rotational Motion

- (1) Applying torque equation about point P.

$$\tau = I \alpha = [2M_0(2\ell)^2 + 5M_0\ell^2]\alpha$$

$$\Rightarrow 5M_0g\ell - 4M_0g\ell = [2M_0(2\ell)^2 + 5M_0\ell^2]\alpha$$

$$\Rightarrow M_0g\ell = (13M_0g\ell^2)\alpha$$

$$\therefore \alpha = \frac{g}{13\ell}$$

10 Jan Evening

Q6

- (1) For Ball

using parallel axes theorem, for ball moment of inertia,

$$I_{\text{ball}} = \frac{2}{5}MR^2 + M(2R)^2 = \frac{22}{5}MR^2$$

For two balls $I_{\text{balls}} = 2 \times \frac{22}{5}MR^2$ and,

$$I_{\text{rod}} = \frac{M(2R)^2}{12} = \frac{MR^2}{3}$$

$$I_{\text{system}} = I_{\text{balls}} + I_{\text{rod}} \\ = \frac{44}{5}MR^2 + \frac{MR^2}{3} = \frac{137}{15}MR^2$$

10 Jan Evening

Q7

JEE Mains 2019 Chapter wise Question Bank

- (1) Let mass of the larger triangle = M

Side of larger triangle = ℓ

Moment of inertia of larger triangle = ma^2

Mass of smaller triangle = $\frac{M}{4}$

Length of smaller triangle = $\frac{\ell}{2}$

Moment of inertia of removed triangle

$$= \frac{M}{4} \left(\frac{a}{2} \right)^2$$

$$\therefore \frac{I_{\text{removed}}}{I_{\text{original}}} = \frac{\frac{M}{4} \left(\frac{a}{2} \right)^2}{M \cdot (a)^2}$$

$$I_{\text{removed}} = \frac{I_0}{16}$$

$$\text{So, } I = I_0 - \frac{I_0}{16} = \frac{15I_0}{16}$$

11 Jan Morning

Q8

Rotational Motion

(1) Given, $\vec{F}_1 = \frac{F}{2}(-\hat{i}) + \frac{F\sqrt{3}}{2}(-\hat{j})$

$$\vec{r}_1 = 0\hat{i} + 6\hat{j}$$

Torque due to F_1 force

$$\vec{\tau}_{F_1} = \vec{r}_1 \times \vec{F}_1 = 6\hat{j} \times \left(\frac{F}{2}(-\hat{i}) + \frac{F\sqrt{3}}{2}(-\hat{j}) \right)$$

$$= 3F(\hat{k})$$

Torque due to F_2 force

$$\vec{\tau}_{F_2} = (2\hat{i} + 3\hat{j}) \times F\hat{k} = 3F\hat{i} + 2F(-\hat{j})$$

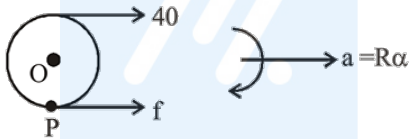
$$\vec{\tau}_{\text{net}} = \vec{\tau}_{F_1} + \vec{\tau}_{F_2} = 3F\hat{i} + 2F(-\hat{j}) + 3F(\hat{k})$$

$$= (3\hat{i} - 2\hat{j} + 3\hat{k})F$$

11 Jan Morning

Q9

(2)



From newton's second law

$$40 + f = m(R\alpha) \quad \dots(i)$$

Taking torque about O we get

$$40 \times R - f \times R = I\alpha$$

$$40 \times R - f \times R = mR^2 \alpha$$

$$40 - f = mR \alpha \quad \dots(ii)$$

Solving equation (i) and (ii)

$$\alpha = \frac{40}{mR} = 16 \text{ rad/s}^2$$

11 Jan Evening

Q10

JEE Mains 2019 Chapter wise Question Bank

(1) Torque about the origin = $\vec{\tau} = \vec{r} \times \vec{F}$

$$= r F \sin \theta \Rightarrow 2.5 = 1 \times 5 \sin \theta$$

$$\sin \theta = 0.5 = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

11 Jan Evening

Q11

(2) Moment of inertia of disc D_1 about $OO' = I_1 = \frac{MR^2}{2}$

M.O.I of D_2 about OO'

$$= I_2 = \frac{1}{2} \left(\frac{MR^2}{2} \right) + MR^2 = \frac{MR^2}{4} + MR^2$$

M.O.I of D_3 about OO'

$$= I_3 = \frac{1}{2} \left(\frac{MR^2}{2} \right) + MR^2 = \frac{MR^2}{4} + MR^2$$

so, resultant M.O.I about OO' is $I = I_1 + I_2 + I_3$

$$\Rightarrow I = \frac{MR^2}{2} + 2 \left(\frac{MR^2}{4} + MR^2 \right)$$

$$= \frac{MR^2}{2} + \frac{MR^2}{2} + 2MR^2$$

$$= 3 MR^2$$

11 Jan Evening

Q12

(2)

12 Jan Morning

Q13

Rotational Motion

- (3) According to work-energy theorem

$$mgh = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2$$

$$2gh = v_B^2 - v_A^2$$

$$2 \times 10 \times 10 = v_B^2 - 5^2$$

$$\Rightarrow v_B = 15 \text{ m/s}$$

Angular momentum about O,

$$L_O = mvr$$

$$= 20 \times 10^{-3} \times 20$$

$$L_O = 6 \text{ kg.m}^2/\text{s}$$

12 Jan Evening

Q14

- (4) According to parallel axes theorem

$$I = \frac{2}{5}mR^2 + mx^2$$

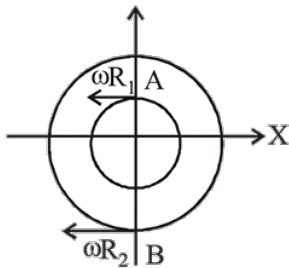
Hence graph (4) correctly depicts I vs x.

12 Jan Evening

Q15

- (3) From, $\theta = \omega t = \frac{\pi}{2\omega} = \frac{\pi}{2}$

So, both have completed quarter circle



Relative velocity,

$$v_A - v_B = \omega R_1(-\hat{i}) - \omega R_2(-\hat{i}) = \omega(R_2 - R_1)\hat{i}$$

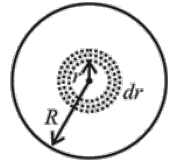
12 Jan Evening

Q16

JEE Mains 2019 Chapter wise Question Bank

- (3) Taking a circular ring of radius r and thickness dr as a mass element, so total mass,

$$M = \int_0^R \rho_0 r \times 2\pi r dr = \frac{2\pi\rho_0 R^3}{3}$$



$$I_C = \int_0^R \rho_0 r \times 2\pi r dr \times r^2 = \frac{2\pi\rho_0 R^5}{5}$$

Using parallel axes theorem

$$\therefore I = I_C + MR^2 = 2\pi\rho_0 R^5 \left(\frac{1}{5} + \frac{1}{3} \right)$$

$$= \frac{16\pi\rho_0 R^5}{15}$$

$$= \frac{8}{5} \left[\frac{2}{3} \pi \rho_0 R^3 \right] R^2 = \frac{8}{5} MR^2$$

8 April Morning

Q17

- (3) For sphere,

$$\frac{1}{2}mv^2 + I\omega^2 = \frac{1}{2}mgh$$

$$\text{or } \frac{1}{2}mv^2 + \frac{1}{2} \left(\frac{2}{5}mR^2 \right) \frac{v^2}{R^2} = mgh \text{ or } h = \frac{7v^2}{10g}$$

For cylinder

$$\frac{1}{2}mv^2 + \frac{1}{2} \left(\frac{mR^2}{2} \right) \frac{v^2}{R^2} = mgh'$$

$$\text{or } h' = \frac{3v^2}{4g}$$

$$\therefore \frac{h}{h'} = \frac{7v^2/10g}{3v^2/4g} = \frac{14}{15}$$

8 April Evening

Q18

(1)

8 April Evening

Q19

- (4) Work done by torque is responsible for change in kinetic energy.

$$\therefore \tau = \frac{dE}{d\theta} \therefore I\alpha = 2K\theta \therefore \alpha = \frac{2K\theta}{I}$$

Rotational Motion

9 April Morning

Q20

$$\begin{aligned} \text{(Bonus)} \quad mgh &= \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2 \\ &= \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\left(\frac{v_{cm}}{R}\right)^2 \\ &= \frac{1}{2}\left(m + \frac{I_{cm}}{R^2}\right)v_{cm}^2 = \frac{1}{2}\left[1 + \frac{K^2}{R^2}\right] \end{aligned}$$

$$\therefore h \propto 1 + \frac{K^2}{R^2}$$

For ring : $h \propto 2$ ($\because K = R$)

$$\text{For solid cylinder, } h \propto \frac{3}{2} \quad \left(\because K = \frac{R}{\sqrt{2}}\right)$$

$$\text{For solid sphere, } h \propto \frac{7}{5} \quad \left(\because K = \sqrt{\frac{2}{5}}R\right)$$

$$\text{Ratio of heights } 2 : \frac{3}{2} : \frac{7}{5} \Rightarrow 20 : 15 : 14$$

9 April Morning

Q21

$$(2) \quad \omega = \alpha t = 20t$$

$$\text{Given, } \frac{1}{2}I\omega^2 = 1200$$

$$\text{or } \frac{1}{2} \times 1.5 \times (20t)^2 = 1200$$

$$\text{or } t = 2 \text{ s}$$

9 April Evening

Q22

$$(3) \quad I_i\omega_i = I_f\omega_f$$

$$\text{or } \left(\frac{ML^2}{12}\right)\omega_0 = \left(\frac{ML^2}{12} + 2m\left(\frac{L}{2}\right)^2\right)\omega_f$$

$$\therefore \omega_f = \left(\frac{M\omega_0}{M + 6m}\right)$$

9 April Evening

Q23

JEE Mains 2019 Chapter wise Question Bank

$$(2) \quad \text{Given that, } x = x_0 + a \cos \omega_1 t$$

$$y = y_0 + b \sin \omega_2 t$$

$$\frac{dx}{dt} = V_x$$

$$\Rightarrow v_x = -a\omega_1 \sin(\omega_1 t), \quad \text{and } \frac{dy}{dt} = v_y = b\omega_2 \cos(\omega_2 t)$$

$$\frac{dv_x}{dt} = a_x = -a\omega_1^2 \cos(\omega_1 t), \quad \frac{dv_y}{dt} = a_y = -b\omega_2^2 \sin(\omega_2 t)$$

$$\text{At } t = 0, x = x_0 + a, y = y_0$$

$$a_x = -a\omega_1^2, a_y = 0$$

$$\text{Now, } \vec{\tau} = \vec{r} \times \vec{F} = m(\vec{r} \times \vec{a}) = [(x_0 + a)\hat{i} + y_0\hat{j}] \times m(-a\omega_1^2\hat{i})$$

$$= +my_0 a \omega_1^2 \hat{k}$$

10 April Morning

Q24

$$(2) \quad \text{As from the question density } (\sigma) = kr^2$$

$$\text{Mass of disc } M = \int_0^R (kr^2) 2\pi r dr = 2\pi k \frac{R^4}{4} = \frac{\pi k R^4}{2}$$

$$\Rightarrow k = \frac{2M}{\pi R^4} \quad \dots(i)$$

$\therefore$ Moment of inertia about the axis of the disc.

$$I = \int dl = \int (dm)r^2 = \int \sigma dA r^2$$

$$= \int (kr^2)(2\pi r dr)r^2$$

$$= \int_0^R 2\pi k r^5 dr = \frac{\pi k R^6}{3} = \frac{\pi \times \left(\frac{2M}{\pi R^4}\right) \times R^6}{3} = \frac{2}{3}MR^2$$

[putting value of k from eqn(i)]

10 April Morning

Q25

Rotational Motion

(4) As no external torque is acting so angular momentum should be conserved

$(I_1 + I_2) \omega = I_1 \omega_1 + I_2 \omega_2$ [ω_c = common angular velocity of the system, when discs are in contact]

$$\omega_c = \frac{I_1 \omega_1 + \frac{I_1 \omega_1}{4} \left(\frac{5}{4} \times \frac{2}{3} \right)}{I_1 + \frac{I_1}{2}} \omega_1$$

$$\omega_c = \frac{5\omega_1}{6}$$

$$E_f - E_i = \frac{1}{2} (I_1 + I_2) \omega_c^2 - \frac{1}{2} I_1 \omega_1^2 - \frac{1}{2} I_2 \omega_2^2$$

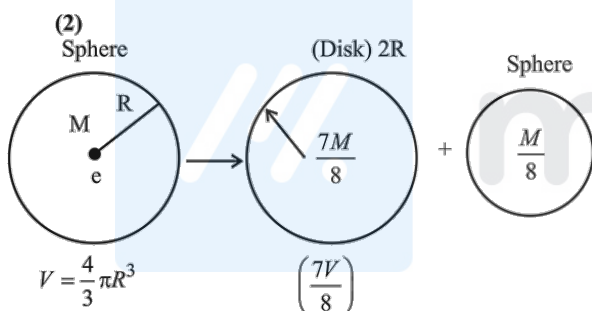
Put $I_2 = I_1/2$ and $\omega_c = \frac{5\omega_1}{6}$ $5\omega_1/6$

We get :

$$E_f - E_i = -\frac{I_1 \omega_1^2}{24}$$

10 April Morning

Q26



$$I_1 = \left(\frac{7M}{8} \right) (2R)^2 \frac{1}{2} = \left(\frac{7}{16} \times 4 \right) MR^2 = \frac{14}{8} mR^2$$

$$I_2 = \frac{2}{5} \left(\frac{M}{8} \right) r^2$$

$$\Rightarrow I_2 = \frac{2}{5} \left(\frac{M}{8} \right) \left(\frac{R^2}{4} \right) = \frac{MR^2}{80}$$

$$\left[\frac{4}{3} \pi r^3 \rho = \frac{1}{8} \frac{4}{3} \pi R^3 \times \rho \right]$$

$$\Rightarrow r = R/2$$

$$\frac{I_1}{I_2} = \frac{14 \times 80}{8} = 140$$

10 April Evening

Q27

JEE Mains 2019 Chapter wise Question Bank

(4) We have given $\vec{r} = 2t\hat{i} - 3t^2\hat{j}$

$$\vec{r}(\text{at } t = 2) = 4\hat{i} - 12\hat{j}$$

$$\text{Velocity, } \vec{v} = \frac{d\vec{r}}{dt} = 2\hat{i} - 6t\hat{j}$$

$$\vec{v}(\text{at } t = 2) = 2\hat{i} - 12\hat{j}$$

$$\vec{L} = mvr \sin \theta \hat{n} = m(\vec{r} \times \vec{v})$$

$$= 2(4\hat{i} - 12\hat{j}) \times (2\hat{i} - 12\hat{j}) = -48\hat{k}$$

10 April Evening

Q28

(4) Angular acceleration,

$$\alpha = \frac{\omega - \omega_0}{t}$$

$$= \frac{25 \times 2\pi - 0}{5}$$

$$= 10 \pi \text{ rad/s}^2$$

$$\tau = I\alpha$$

$$\Rightarrow \tau = \left(\frac{5}{4} mR^2 \right) \alpha$$

$$\approx \left(\frac{5}{4} \right) (5 \times 10^{-3}) (10^{-4}) 10\pi$$

$$= 2.0 \times 10^{-5} \text{ Nm}$$

10 April Evening

Q29

$$\begin{aligned}
 (3) \quad I &= \int_a^b (dm)r^2 \\
 &= \int_a^b \left(\frac{\sigma_0}{r} \times 2\pi r dr \right) r^2 \\
 &= \frac{2\pi\sigma_0}{3} \left| r^3 \right|_a^b \\
 &= \frac{2\pi\sigma_0}{3} (b^3 - a^3)
 \end{aligned}$$

Mass of the disc,

$$m = \int_a^b \frac{\sigma_0}{r} \times 2\pi r dr = 2\pi\sigma_0 (b - a)$$

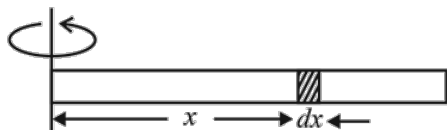
Radius of gyration,

$$\begin{aligned}
 k &= \sqrt{\frac{I}{m}} \\
 &= \sqrt{\frac{(2\pi\sigma_0/3)(b^3 - a^3)}{2\pi\sigma_0(b - a)}} \\
 &= \sqrt{\frac{a^2 + b^2 + ab}{3}}
 \end{aligned}$$

12 April Morning

Q30

$$(4) \quad \int_0^T (-dT) = \int_l^x (dm)\omega^2 x$$



$$-T = \int_l^x \left(\frac{m}{l} dx \right) \omega^2 x$$

$$\text{or } T = \frac{m\omega^2}{l} (l^2 - x^2)$$

It is a parabola between T and x .

12 April Morning